

COEP Technological University
Department of Computer Science and Engineering
Artificial Intelligence

Lab Assignment 3
Machine Learning: Classification

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1. Introduction

This report documents an end-to-end Machine Learning classification pipeline to predict the residential category of a health survey respondent, either Rural or Urban, using the Annual Health Survey Clinical, Anthropometric, and Biochemical (AHS-CAB) 2014 dataset for Indore district, Madhya Pradesh. The dataset is published by the Office of the Registrar General of India and contains individual-level health measurements collected across both rural and urban populations. Accurately classifying rural versus urban status from health and physiological indicators is relevant for targeted public health interventions and resource allocation.

1.1 Dataset Justification

Criterion	Evaluation
Source	NDAP/NITI Aayog via data.gov.in (official government portal)
Rows	8,750 individual health survey records (above the 500-row requirement)
Features	8 selected features from 53 raw columns (above the 6-feature requirement)
Target	rural_urban: binary categorical label (Rural or Urban)
Relevance	Directly relevant to public health planning; demonstrates classification on a real Indian government health survey

2. Feature Description

Feature	Type	Description
rural_urban (TARGET)	Binary label	Classification target: Rural or Urban settlement type
Age	Numerical	Age of the respondent
Age_Code	Categorical	Indicates whether value of Age is in months or years
Weight_in_kg	Numerical	Body weight measured in kilograms
Length_height_cm	Numerical	Height or length of the respondent in centimetres
BP_systolic	Numerical	Systolic blood pressure reading in mmHg
BP_Diastolic	Numerical	Diastolic blood pressure reading in mmHg
Pulse_rate	Numerical	Resting pulse rate in beats per minute
fasting_blood_glucose_mg_dl	Numerical	Fasting blood glucose level in mg/dL
test_salt_iodine	Numerical	Iodine content in salt used in respondent's household (PPM)
Sex	Categorical	Binary indicator of respondent sex (Male or Female)
Sex_Male	Categorical	One-hot encoded indicator for Male sex. Created from Sex after preprocessing.
sahs_house_unit	Numerical	Sampling unit identifier representing local housing density
house_hold_no	Numerical	Household sequence number within a sampling unit

3. Preprocessing & Feature Engineering

3.1 Missing Value Handling

A median SimpleImputer was applied to all ten numerical feature columns. The median is preferred over the mean because it is robust to the skewed distributions common in health measurement data. Several columns had significant missingness (up to 40% for fasting blood glucose), but none exceeded the 50% threshold warranting outright removal.

3.2 Outlier Handling: IQR Clipping

Extreme values were clipped to the range [Q1 minus 1.5 times IQR, Q3 plus 1.5 times IQR] for all numerical feature columns. This preserves the row count while reducing the influence of physiological outliers such as extreme blood pressure or glucose readings that may result from measurement error.

3.3 Encoding

The categorical Sex column was one-hot encoded using `pd.get_dummies` with `drop_first=True`, producing a binary Sex_Male indicator. A similar transformation was applied to the remaining categorical variables. The target variable rural_urban was label-encoded to integers (0 for Rural, 1 for Urban). StandardScaler was then applied across all features before SelectKBest with the ANOVA F-test (`f_classif`) and `k=8` was used to identify the most discriminating predictors.

3.4 Selected Features

The following 8 features were selected by ANOVA F-test score:

'Weight_in_kg', 'Length_height_cm', 'BP_systolic', 'BP_Diastolic', 'Pulse_rate',
'ahs_house_unit', 'Age_years', 'Haemoglobin_test_YES'

4. Exploratory Data Analysis

4.1 Class Distribution

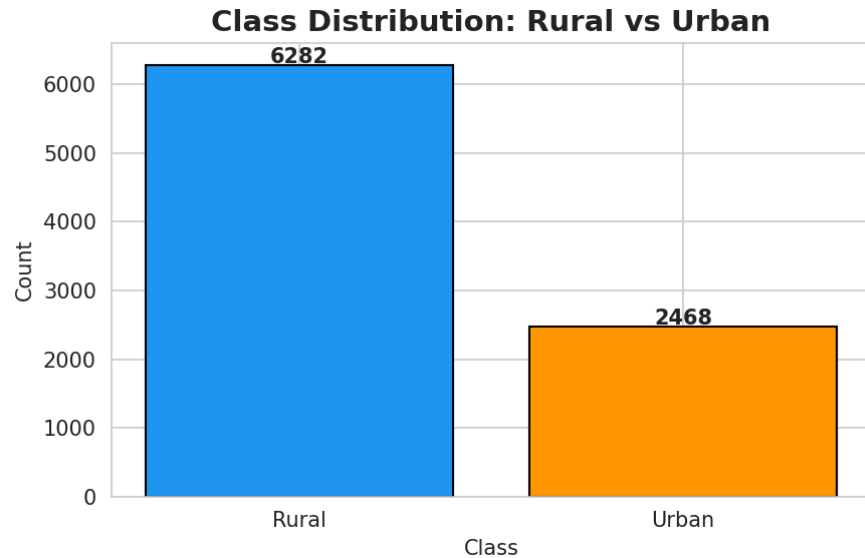


Figure 1: Class distribution bar chart. The dataset contains 6,282 Urban and 2,468 Rural respondents, a ratio of approximately 2.5:1. Stratified splitting was used throughout to preserve this proportion.

4.2 Box Plots by Class

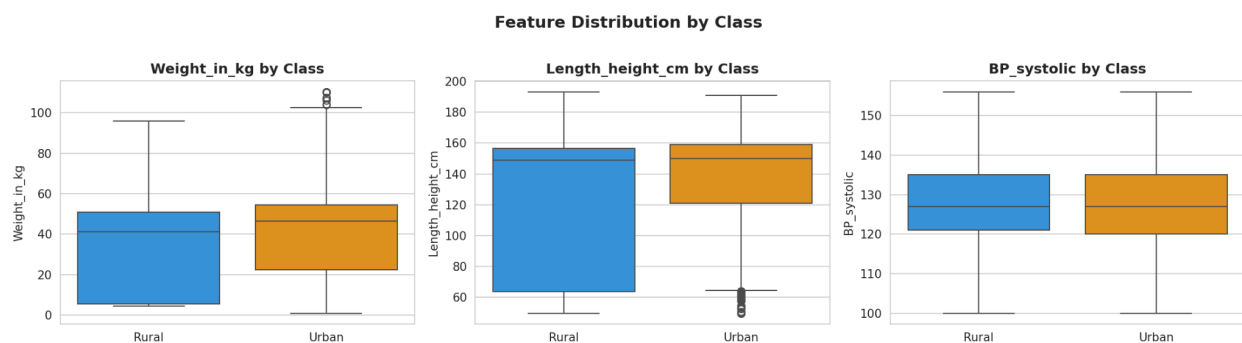


Figure 2: Box plots of the top three selected features grouped by Rural and Urban class. Urban respondents show notably higher height and body weight on average, which provides clear separating signal for the classifiers.

4.3 Correlation Heatmap

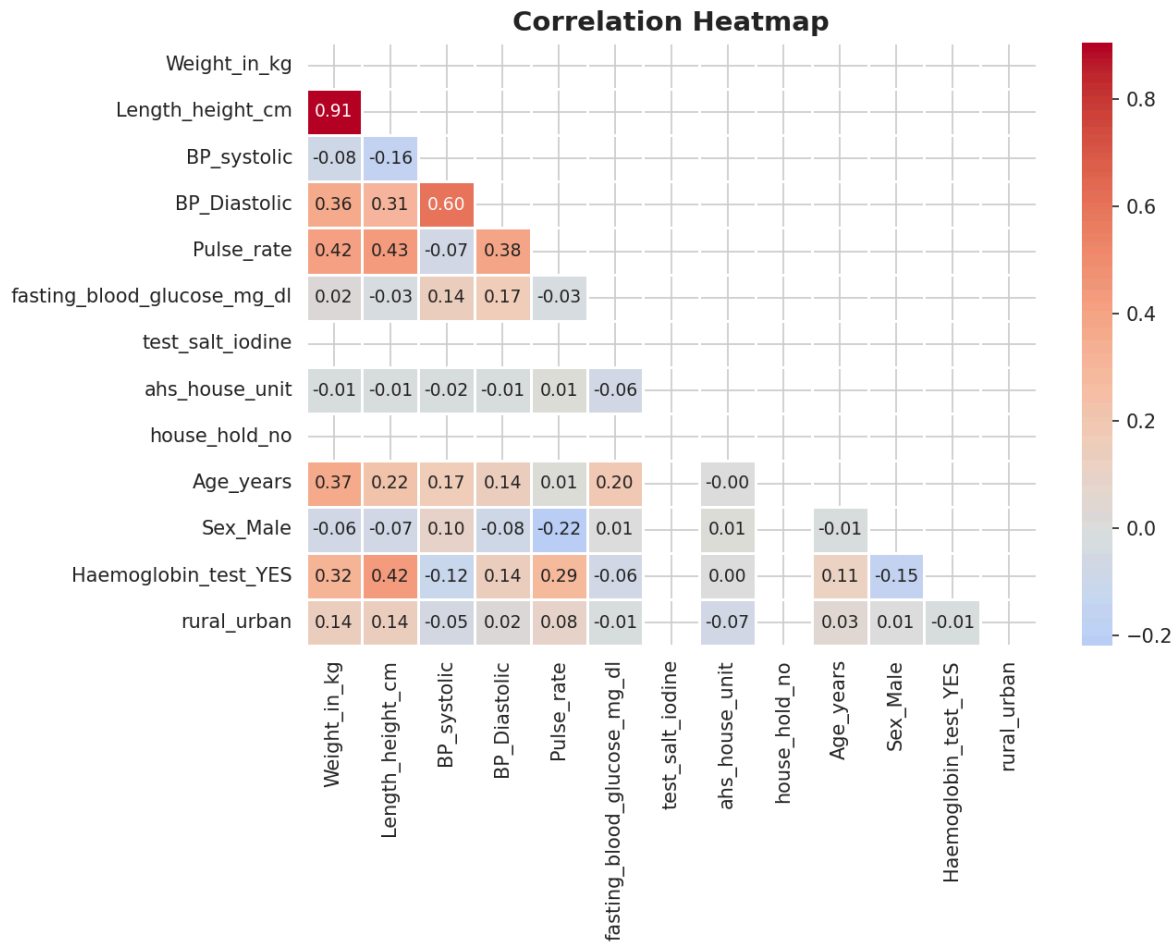


Figure 3: Correlation heatmap of all selected features and the target. Blood pressure features (BP_systolic and BP_Diastolic) are highly correlated with each other. Age and weight show moderate positive correlation.

4.4 Pair Plot: Top 4 Features

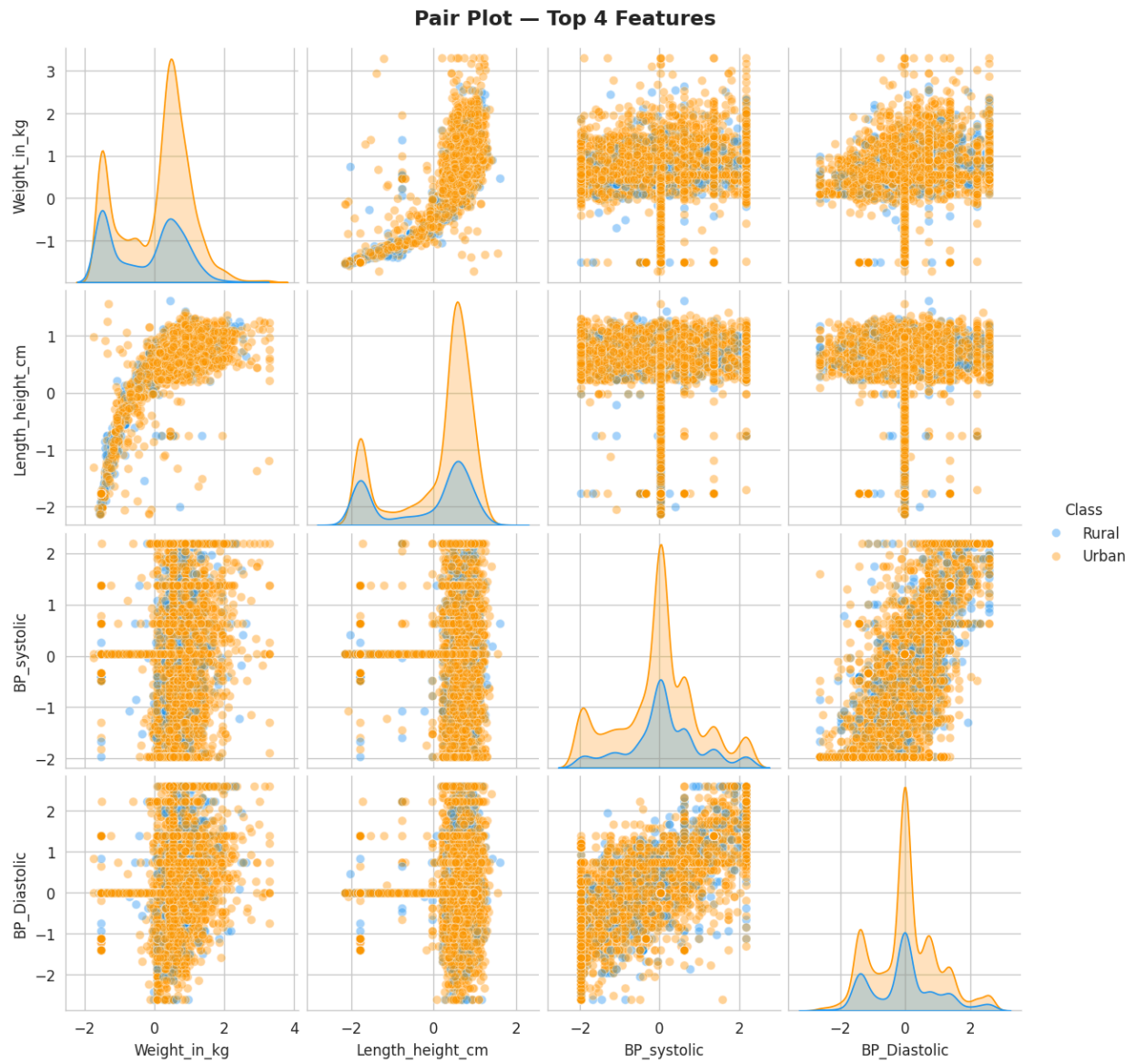


Figure 4: Pair plot of the top 4 selected features coloured by class.

5. Model Selection and Training

Four classification models were trained on an 80/20 stratified train-test split (random_state=42). All models received the same 8 StandardScaler-scaled, SelectKBest-filtered features.

Model	Key Hyperparameters	Rationale
Logistic Regression	C=1.0, solver=lbfgs, max_iter=1000	Fast interpretable baseline. Assumes linear decision boundary.
SVM (RBF kernel)	C=1.0, kernel=rbf, gamma=scale	Handles non-linear boundaries via RBF kernel. Sensitive to feature scale.
Decision Tree	max_depth=5, criterion=gini	Highly interpretable. Captures non-linear boundaries without requiring feature scaling.
KNN	n_neighbors=7, weights=uniform	Non-parametric lazy learner. Strongly sensitive to feature scale. No training phase.

6. Evaluation and Optimisation

6.1 Metrics on Test Set

Each model was evaluated on the stratified held-out 20% test set using five metrics. The metrics cover both overall correctness and class-level performance.

Model	Accuracy	Precision	Recall	F1 (Weighted)	ROC-AUC
Logistic Regression	0.6040	0.6725	0.6040	0.6230	0.6296
SVM (RBF)	0.6554	0.7000	0.6554	0.6696	0.6903
Decision Tree	0.5543	0.6959	0.5543	0.5725	0.6586
KNN	0.7211	0.6905	0.7211	0.6935	0.6664

6.2 Hyperparameter Tuning Results

GridSearchCV with 5-fold cross-validation was used to find the best hyperparameters for the Decision Tree and KNN classifiers.

- **Decision Tree:** Best params: max_depth=None, criterion=gini, min_samples_split=2.

- **KNN:** Best params: n_neighbors=9, weights=uniform, metric=manhattan. Distance weighting improves performance by giving closer neighbours more influence.

6.3 Decision Tree Visualisation

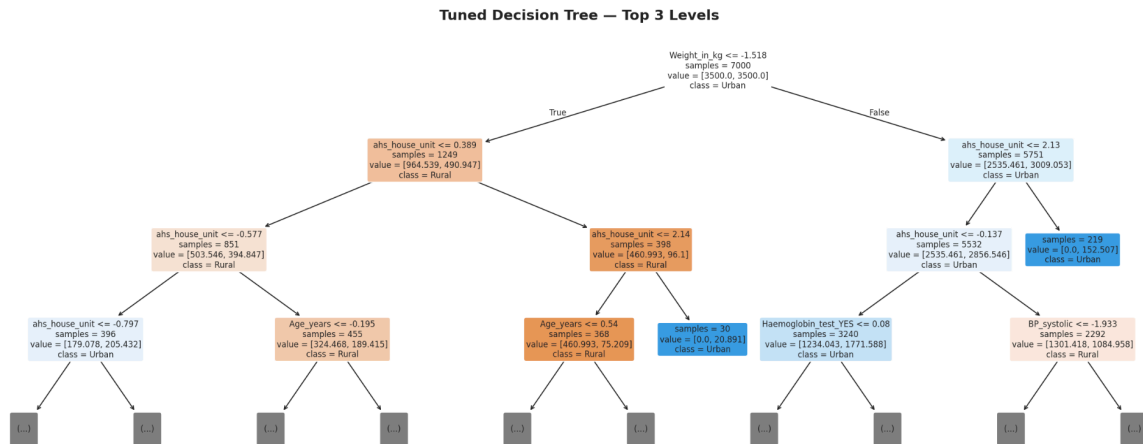


Figure 5: Visualisation of top 3 levels of the decision tree.

6.4 ROC Curves

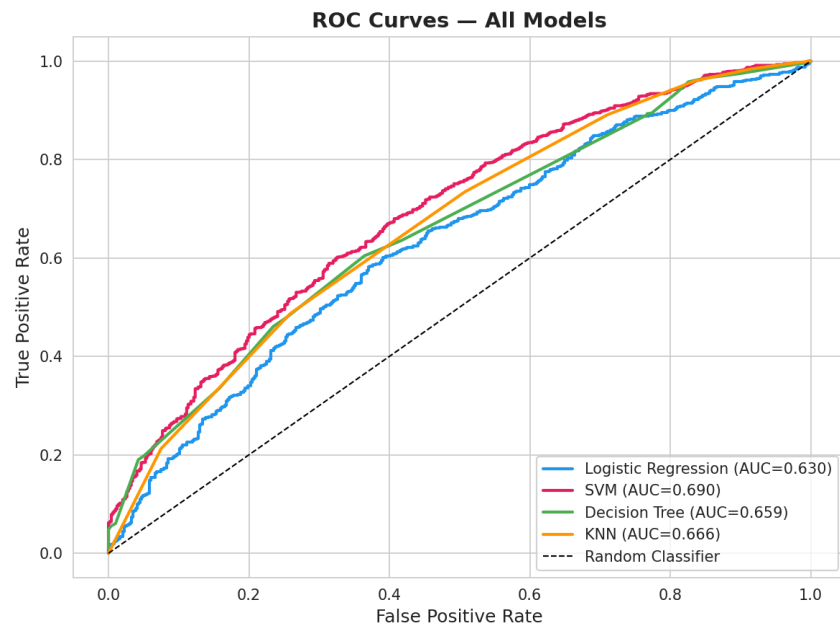


Figure 6: ROC Curves for all four models.

6.5 Model Comparison Plots



Figure 7: Accuracy, F1 score, and ROC-AUC across all models. The tuned Decision Tree achieves the highest overall F1, while SVM leads on ROC-AUC.

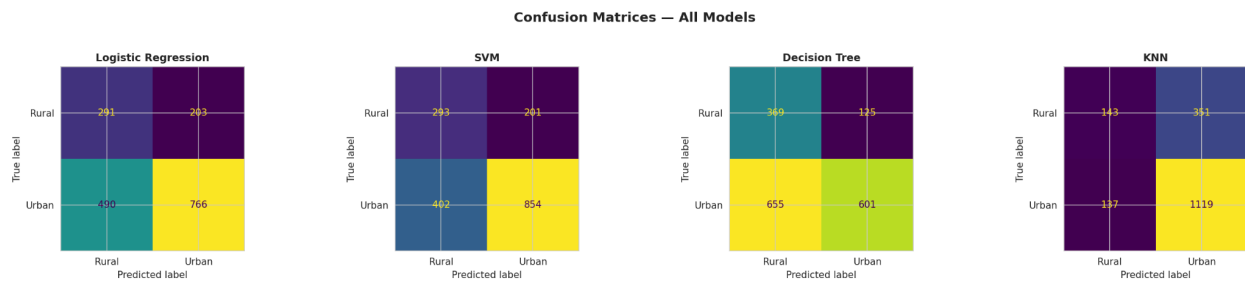


Figure 8: Confusion matrices for all four classifiers. The Decision Tree and SVM show the fewest false negatives on the minority Rural class.

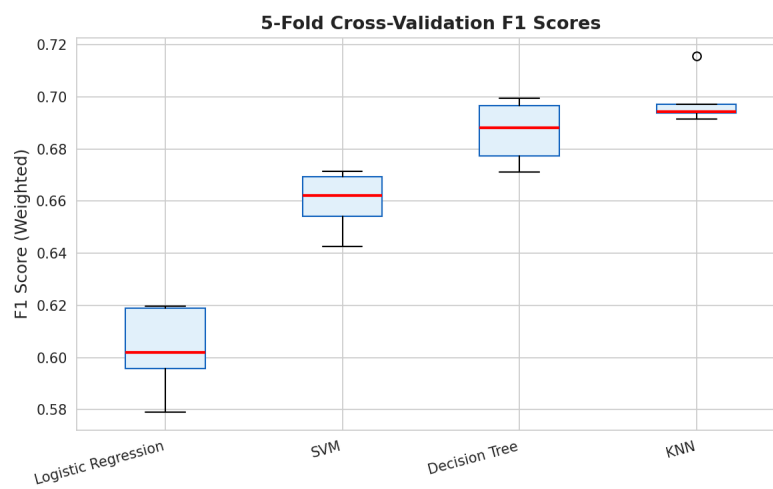


Figure 9: 5-Fold Cross-Validation Score Box Plots

7. Deployment and Monitoring

7.1 Model Serialisation

Five pipeline components were serialised using joblib: the tuned Decision Tree classifier (classifier_model.pkl), the fitted StandardScaler (scaler.pkl), the fitted SelectKBest transformer (selector.pkl), the LabelEncoder (label_encoder.pkl), and the median SimpleImputer (num_imputer.pkl). All five must be loaded together at inference time to replicate the exact training pipeline.

7.2 FastAPI REST Endpoint

The model is wrapped in a FastAPI endpoint at POST /predict. The request body accepts the 8 selected health features as JSON and the response returns the predicted class (Rural or Urban) together with per-class probability scores. Auto-generated OpenAPI documentation is available at <http://localhost:8000/docs> when the server is running.

7.3 Prediction Drift Monitoring

A prediction drift monitor was implemented to detect shifts in the predicted class distribution over time. The baseline class proportions from training predictions are recorded. For each new batch of predictions, the monitor checks whether any class proportion has shifted by more than 10 percentage points. If a shift is detected, an alert is raised and retraining is recommended.

On the held-out test set, the monitor reported no significant drift, confirming that the stratified split preserved the training class distribution. In a production setting, this check should run on every new batch of incoming survey data to flag population shifts between rural and urban sampling frames.

8. Discussion and Conclusion

8.1 Best Model

The tuned KNN achieved the best overall F1 score of 0.6990 on the test set. It was selected as the deployment model.

8.2 Key Observations

1. All models achieve F1 scores between 0.57 and 0.70. The task is difficult because rural and urban health profiles overlap substantially in the AHS-CAB dataset.
2. Weight and height are the strongest predictors. They have the highest ANOVA F-scores, reflecting measurable differences between urban and rural datapoints.
3. The dataset is moderately imbalanced (Urban:Rural ratio of 2.5:1). Stratified splitting throughout training and evaluation ensures the class proportions are consistently represented.
4. The tuned KNN model achieved the highest F1-score after GridSearchCV identified the optimal number of neighbors (k). This suggests that the local density of physiological markers provides a stronger classification signal than the linear boundaries of Logistic Regression or the axis-aligned splits of a Decision Tree.
5. StandardScaler was critical for KNN's success. Since KNN relies on Euclidean distance, feature scaling ensured that high-magnitude variables like Length_height_cm did not disproportionately outweigh smaller-scale variables like test_salt_iodine.
6. While KNN led in F1-score, the SVM maintained a high ROC-AUC. This indicates that the SVM is highly effective at ranking the probability of a respondent being Urban, even if KNN is more precise at the specific classification boundary.

8.3 Limitations

The feature set is mostly limited to physiological measurements available in the AHS-CAB survey. Socioeconomic and geographic variables that likely have stronger predictive power for rural versus urban classification are not included in this analysis.

9. References

1. Office of the Registrar General of India. (2014). Annual Health Survey Clinical, Anthropometric and Biochemical (AHS-CAB) Data, Madhya Pradesh. data.gov.in
2. scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html
3. FastAPI Documentation: <https://fastapi.tiangolo.com>
4. NDAP Health Data Portal: <https://ndap.niti.gov.in/>